CSAI 202

Introduction to Database Systems



Welcome



RAW

Ahmed is student 101 in Computer Science. His GPA is 3.2.

Sara is student 102 in Computer Science. Her GPA is 3.7.

Omar is student 103 in Artificial Intelligence. His GPA is 3.1.

Sara is also listed as student 102 in CS, with GPA 3.7. VIOLATION

Ali is student 104 in CS, with GPA −5. VIOLATION

Mona is student 105, but her major is missing. VIOLATION

What is wrong? Nothing tells the system what a valid student record should look like.

File (csv)

StudentID	Name	Major	GPA	
101	Ahmed	CS	3.2	
102	Sara	CS	3.7	
103	Omar	AI	3.1	
102	Sara	Computer Science	3.7	duplicate ID + inconsistent major
104	Ali	CS	-5	invalid GPA
105	Mona		3.5	missing major

A file can store the values exactly as entered. It does not automatically enforce the meaning of the values.

Table

StudentID	Name	Major	GPA	Problem
101	Ahmed	CS	3.2	OK
102	Sara	CS	3.7	OK
103	Omar	AI	3.1	OK
102	Sara	Computer Science	3.7	Duplicate ID / inconsistent major
104	Ali	CS	-5	Invalid GPA
105	Mona	–	3.5	Missing major

A table improves organization – but the database still needs rules about what is allowed.

Constraints

PRIMARY KEY

StudentID must be unique and present.

Rule

NOT NULL

Name and MajorID are required.

Rule

CHECK

GPA must be between 0 and 4.

Rule

FOREIGN KEY

MajorID must refer to a valid major.

Rule

Now the database can reject bad data instead of merely storing it.

Database

STUDENT

StudentID	Name	MajorID	GPA
101	Ahmed	CS	3.2
102	Sara	CS	3.7
103	Omar	AI	3.1

MAJOR

MajorID	MajorName
CS	Computer Science
AI	Artificial Intelligence
IS	Information Systems



The database rejects:

- StudentID = 102 again → PRIMARY KEY violation
- GPA = −5 → CHECK violation
- MajorID = “XYZ” → FOREIGN KEY violation

Result: clean, consistent, connected data.

Contact Info

- Yousry AbdulAzeem
  - ★ Associate Professor, Computer Engineering
  - ★ Email: yabdulazeem@zewailcity.edu.eg
- Office hours:
  - ★ Sunday 11→14
  - ★ Monday 12→15
  - ★ Tuesday: 9-10 & 12-14

Requirements for this Class

- You are proficient in a programming language.
- You want to learn more about the technical aspects of analysis and design of database systems.

Overall Aims of Course

- Know the ***basic concepts*** of database systems
- Design and practice creating database systems using the ***entity-relationship*** model.
- Learn ***functional dependencies*** and practice relational model ***normalization***
- Learn and practice using **SQL** for relational databases programming.
- Learn and practice using ***relational algebra*** for query optimization.

Readings

- **TEXTBOOK**
  - [1] Ramez Elmasri and Shamkant B. Navathe, ***Fundamentals of Database Systems***, 7th Edition, 2015, Addison-Wesley, ISBN 0-321-36957-2.
- REFERENCES
  - [1] C. J. Date, ***An Introduction to Database Systems***, 8th Edition, 2003, Addison-Wesley, ISBN 0-321-19784-4.
  - [2] Thomas Connolly and Carolyn Begg, ***Database Systems***, 4th Edition, 2005, Pearson Education, ISBN 0-273-70413-3.
- Additional books may be recommended during individuals lectures

Topics We Will Cover



Basic Concepts



Data Modeling



Relational Data Model



Relational Database Design



Functional Dependency and Normalization



Relational Algebra & query optimization



SQL and Database Programming



File Structures & Hashing Techniques

Assessment Strategy

Assessment Method	Degrees
Assessments (best 3 of 4) (week 3, 6, 9, 12)	15
Midterm Exam (week 8)	20
Lecture Participation	5
Lab Assignments	5
Term Project	15
Final Exam	40
Total	100



Detailed Syllabus

W	Lecture	Lab	Assessments	Project
1	· Introduction to Database systems	· Introduction to C# programming -1		
2	· Overview of Database languages and architectures	· Introduction to C# programming -2 (TA)		· Practice C# previous knowledge · TAs Send the project Document to the Students
3	· Data Models (ER Diagram)	· Sheet (DB introduction) · C# Practice session	Assessment 1	· Project Proposal (***3 different ideas***)
4	• Relational DB design	• Sheet (ER, DB Constraints)		· Proposal Feedback
5	· Relational DB design Cont. · Mapping a conceptual design into a logical design	· Sheet (Relational Mapping)		· ER Diagram Delivery (initial version)
6	· Database Programming: SQL	· Introduction to SQL Server	Assessment 2	· ER Feedback
7	**Midterm**			

Detailed Syllabus

W	Lecture	Lab	Quizzes	Project
8	· Advanced SQL	· Sheet (Basic SQL) · Accessing DB from C# (cont.)		· ER Diagram Delivery (Final) after modifying the TAs comments
9	· Advanced SQL	· Sheet (More SQL)	Assessment 3	Database Schema
10	· FD introduction · Normalization	· Accessing DB from C# App	**Quiz2 (SQL)**	
11	· Revision :FDs	· Sheet (FD Normalization) · Stored Procedures, Triggers		
12	· Relational Algebra Intro	· Crystal Reports	Assessment 4	
13	· Algebra cont. · Query Optimization	· Algebra Sheet		· Project Delivery
14	Project discussion		**Quiz3 (FD)**	

Thank you

Questions?

